

SIXTH SEMESTER DIPLOMA EXMINATION IN MECHANICAL ENGINEERING**INDUSTRIAL REFRIGERATION****MODEL QUESTION****(Note:Use of Psychrometric Chart, Refrigerant Charts and Tables are permitted)**

(Time : 3 hours)

(Maximum marks : 100)

PART – A**(Maximum Marks :10)**

Marks

I. Answer the following questions in one or two sentences. Each question carries 2 marks

1. State any four desirable properties of an ideal refrigerant
2. State the advantages of multi stage compression
3. State the functions of expansion devices
4. Name any two materials used for piping
5. State the causes of food spoilage (Any Four) (5x2=10)

PART-B

II. Answer any five of the following. Each question carries 6 marks

1. Draw the flow diagram and p-H diagram of the simple vapour compression refrigeration system
2. Explain the working of a vane compressor with sketch
3. Give brief description of capillary tube
4. State the various components of latent heat load
5. Explain cold storage of food preservation
6. Give brief description of solenoid valve
7. Explain cryogenic refrigeration system (5x6=30)

PART-C**Marks****(Maximum Marks : 60)**

III. Answer ONE full question from each module. Each question carries 15 marks

Module –I

- a.Explain Electrolux Refrigeration System with Sketch (8)
- b.Explain solar Refrigeration with sketch (7)

OR

IV

- a. Calculate The power needed to compress 20 kg/minute of R-22 from saturated vapour at 1.4 bar to a condensing pressure of 10 bar by two stage compression with inter cooling by liquid refrigerant at 4 bar. Assume saturated liquid to leave the condenser and dry saturated vapours to leave the evaporator (8)
- b. Explain the process of flash gas removal (7)

Module-II

V

- a. With the aid of sketch explain a surface condenser (8)
 - b. Give a brief description of cooling towers (7)
- OR

VI

- a. Explain reverse cycle defrosting with sketch (8)
- b. Give a brief description about purging (7)

Module-III

VII

- a. The following data relates to the office airconditioning plant having seating capacity of 25 occupants
 - Outside design conditions : 34⁰ C DBT, 28⁰ C WBT
 - Inside design Conditions : 24⁰ C DBT, 50% RH
 - Solar Heat Gain : 9120 W
 - Latent Heat Gain per person : 105 W
 - Sensible Heat Gain per Person : 90W
 - Lighting Load : 2300 W
 - Sensible heat from other sources : 11630 W
 - Infiltration load : 14 m³/minuteFind the Room Sensible Heat Factor (8)
 - b. Explain capacity control of Reciprocating compressors (7)
- OR

VIII

- a. Explain Pressure relief valve and the CSP valve (8)
- b. State the various methods to reduce heat loads (7)

Module-IV

IX

- a. Explain Boot Strap Air cooling of Aircrafts with sketch (8)
 - b. Explain Unitary airconditioning system (7)
- OR

X

- a. Explain the production of dry ice with sketches (8)
- b. Give a brief description about evaporative cooling (7)

